

Data Structures and Algorithms - Unit 1

Test Notes for StudyMate AI

1. Data Structure

A data structure is a way of organizing and storing data so that it can be accessed and modified efficiently. Common data structures include arrays, linked lists, stacks, queues, trees, and graphs.

2. Algorithm

An algorithm is a finite sequence of clear steps used to solve a problem. A good algorithm should be correct, unambiguous, finite, and efficient.

3. Time Complexity

Time complexity describes how the running time of an algorithm grows as the input size increases. Big O notation is commonly used to express the upper-bound growth rate. Examples include $O(1)$, $O(\log n)$, $O(n)$, $O(n \log n)$, and $O(n^2)$.

4. Array

An array stores elements of the same data type in contiguous memory locations. Each element is accessed using an index. Array access by index is generally $O(1)$, while inserting or deleting an element in the middle may require shifting elements and can take $O(n)$ time.

5. Linked List

A linked list is a collection of nodes where each node contains data and a link to another node. In a singly linked list, each node points to the next node. Linked lists allow dynamic memory allocation and efficient insertion or deletion when the required position is known.

6. Stack

A stack follows the LIFO principle: Last In, First Out. The main operations are push, pop, and peek. Stacks are used in function calls, expression evaluation, undo operations, and backtracking.

7. Queue

A queue follows the FIFO principle: First In, First Out. The main operations are enqueue and dequeue. Queues are commonly used in scheduling, buffering, and breadth-first search.

8. Important Exam Points

- Array: contiguous memory and index-based access.
- Linked list: nodes connected using links.
- Stack: LIFO.
- Queue: FIFO.
- Big O: describes algorithm growth with input size.
- Choose a data structure according to the operations required.